



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VETRIVEL AND CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

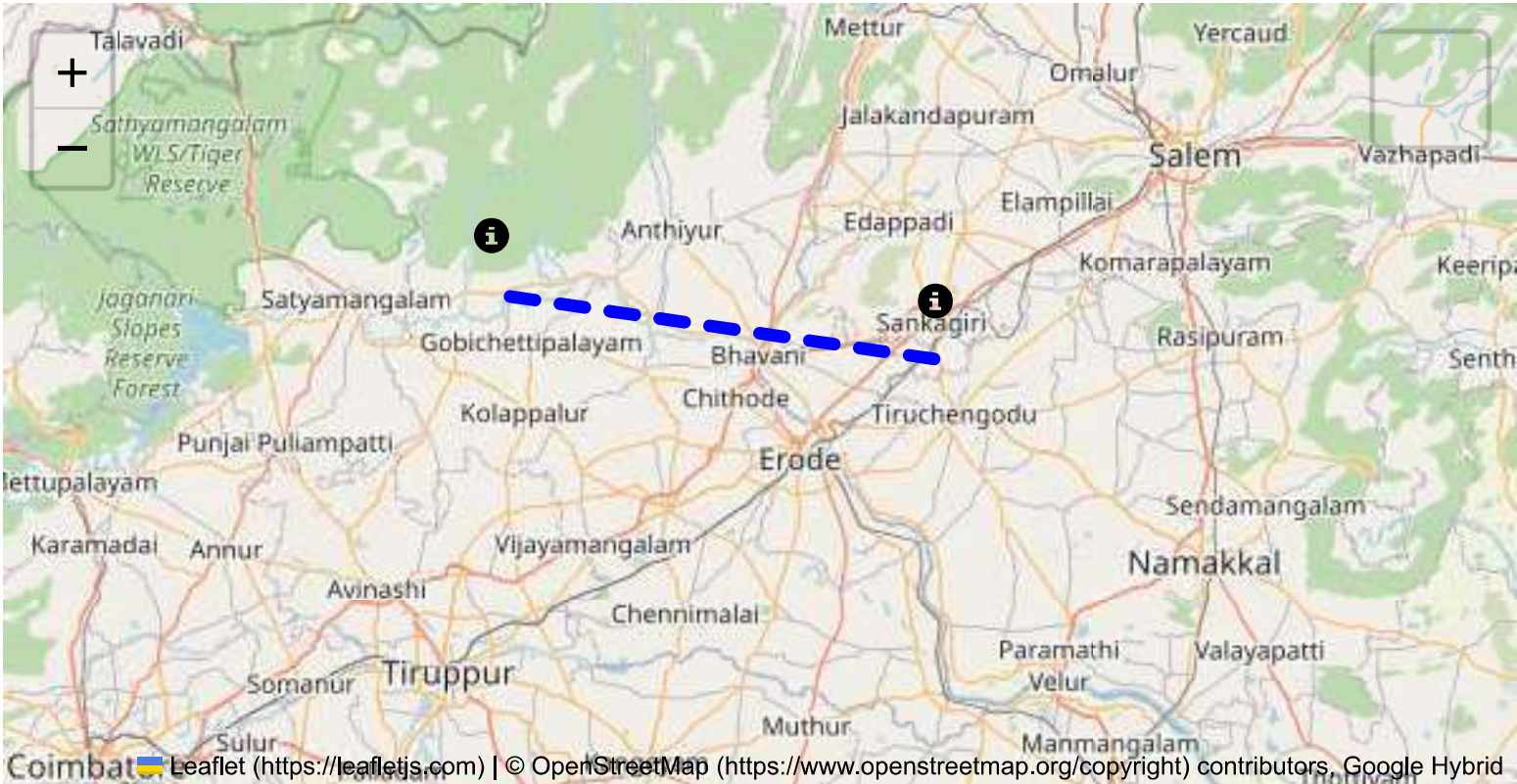
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

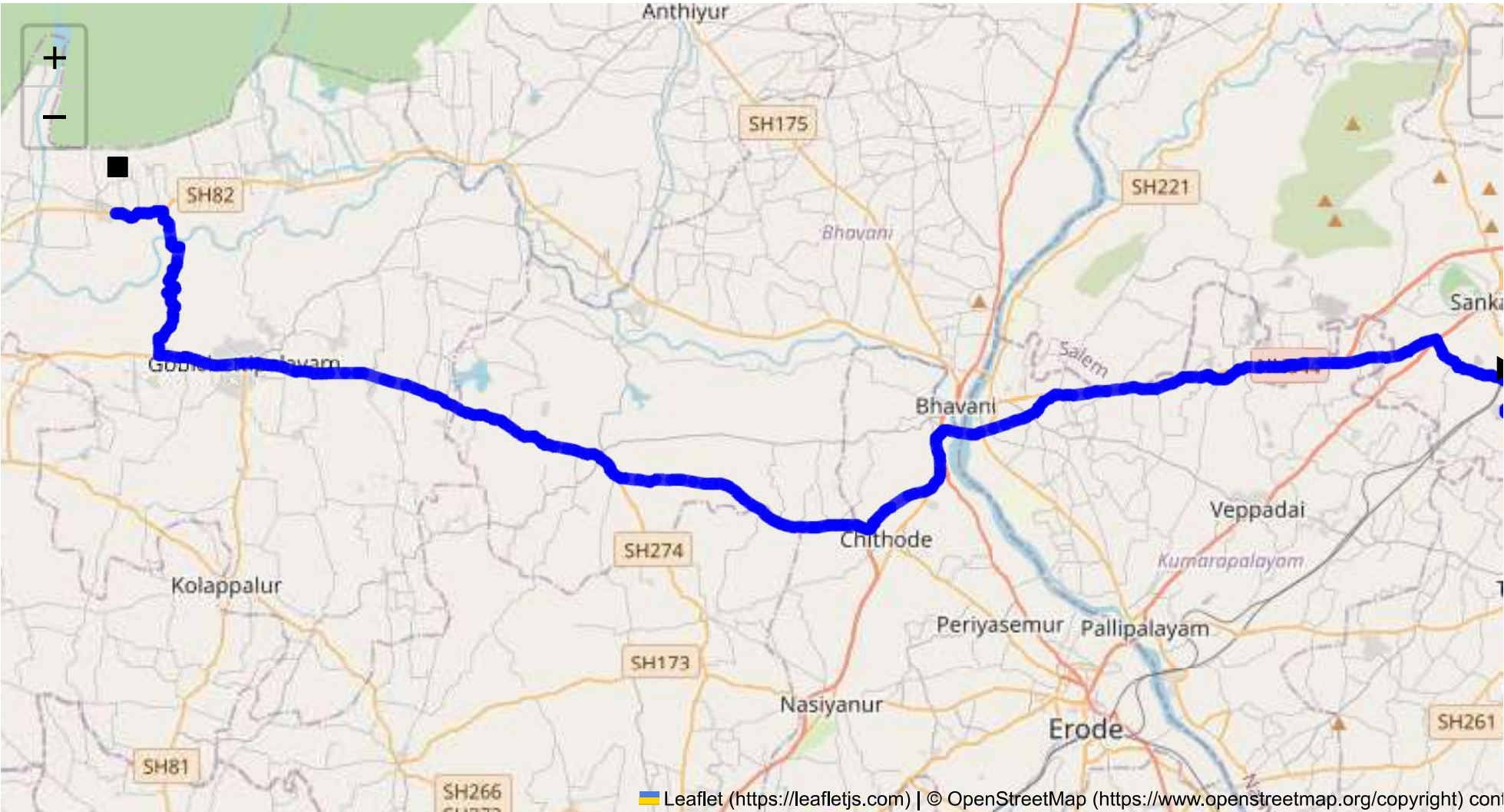
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 66.62 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.50661, 77.390397)



Welcome to the Journey Risk Management Study

Route Safety Analysis: CVQF+23W, Sangagiri to G94R+M34, Tukkanayakkanpalaiyam

1. Overview of the Route Map

The route from Sangagiri to Tukkanayakkanpalaiyam spans approximately 66.62 kilometers, primarily along state highways and local roads. The journey cuts through rural areas with agricultural landscapes, small towns, and a few industrial zones.

2. Weather Conditions and Potential Hazards

- **Typical Weather:** Tamil Nadu generally experiences a tropical climate with hot summers, moderate monsoons, and mild winters.
- **Weather-Related Hazards:** During the monsoon season (June to November), heavy rain can lead to road flooding and reduced visibility. In the summer months, high temperatures may affect road conditions and vehicle performance.

3. Traffic Patterns

- **Peak Hours:** Congestion typically occurs during morning (7-10 AM) and evening (5-8 PM) rush hours, especially near market areas and when crossing towns.
- **Congestion Areas:** Road sections near town centers and markets may witness increased traffic density. Intersections with poorly regulated traffic lights or signage also pose risks.

4. Road Quality and Infrastructure

- **Quality:** Generally, the highways are paved but may have uneven surfaces, potholes, and narrow segments, especially in rural stretches.
- **Infrastructure:** Limited rest stops and service stations are present; the condition varies significantly between urban and rural segments.

5. Alternative Routes for Emergencies

- **Southern Alternative:** Taking a southern route via Erode may provide detours in case of blockage or emergency but could add to the travel time.
- **Northern Route:** Deviating towards Salem and then rejoining the main route; also useful in emergencies but similar in terms of added distance.

6. Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to specific regulations including obtaining proper permits. Restricted hours for transport through populated areas may apply.

7. Historical Incidents

- **Industries and Historical Risks:** The region has seen incidents involving chemical spills and minor collisions involving heavy vehicles, often due to inadequate road conditions or weather factors.
- **Prevention:** Enhanced safety protocols and training sessions have been implemented in response to past incidents.

8. Environmental Considerations and Sensitive Areas

- Protected Areas:** The route passes near agricultural lands where environmental sensitivity is high; precautions during any potential spill are critical.
- Wildlife Crossings:** Be alert for wildlife crossings in rural areas.

9. Communication Coverage

- Network Coverage:** Major towns along the route have good telecommunications coverage. Rural stretches may experience occasional dead zones, especially in hilly or remote areas.

10. Emergency Response Times

- Urban Segments:** Estimated response time is approximately 20-30 minutes due to better access and facilities.
- Rural Segments:** Response times can extend up to 45 minutes to an hour due to limited accessibility and facilities.

11. Overall Summary of Risk Assessment

- Primary Risks:** Include weather conditions during monsoons, road quality variations, and congestion near towns.
- Recommendations:** Schedule trips avoiding peak traffic hours and adverse weather conditions. Ensure communication devices are functional and accessible at all times to mitigate dead zone risks.

In conclusion, while manageable with proper planning and adherence to local regulations, the route demands caution due to potential environmental and road infrastructure challenges. Regular consultations with updated local transport policies and an established emergency plan will contribute to safer transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	Medium	11.45509, 77.81383	30 KM/Hr
10	Turn	High	11.40046, 77.65371	15 KM/Hr
11	Turn	High	11.40035, 77.65383	15 KM/Hr
12	Turn	High	11.39829, 77.65233	15 KM/Hr
13	Turn	Medium	11.42079, 77.56127	30 KM/Hr
14	Turn	High	11.45790, 77.40551	15 KM/Hr
15	Turn	Medium	11.46345, 77.40571	30 KM/Hr
16	Turn	Medium	11.46390, 77.40581	30 KM/Hr
17	Turn	Medium	11.46411, 77.40610	30 KM/Hr
18	Turn	High	11.46407, 77.40677	15 KM/Hr
19	Turn	Medium	11.46630, 77.40811	30 KM/Hr
20	Turn	Medium	11.47136, 77.41058	30 KM/Hr
21	Turn	Medium	11.47169, 77.41050	30 KM/Hr
22	Turn	Medium	11.47207, 77.41009	30 KM/Hr
23	Turn	Medium	11.47242, 77.41006	30 KM/Hr
24	Turn	Medium	11.47535, 77.41069	30 KM/Hr
25	Turn	Medium	11.47796, 77.40971	30 KM/Hr
26	Turn	Medium	11.47858, 77.40888	30 KM/Hr
27	Turn	Medium	11.47910, 77.40859	30 KM/Hr
28	Turn	High	11.47932, 77.40859	15 KM/Hr
29	Turn	Medium	11.48060, 77.41086	30 KM/Hr
30	Turn	Medium	11.48081, 77.41092	30 KM/Hr
31	Turn	Medium	11.48178, 77.40992	30 KM/Hr
32	Turn	High	11.48370, 77.40914	15 KM/Hr
33	Turn	Medium	11.48441, 77.40933	30 KM/Hr
34	Turn	Medium	11.48861, 77.41164	30 KM/Hr
35	Turn	High	11.49474, 77.41231	15 KM/Hr
36	Turn	Medium	11.49523, 77.40994	30 KM/Hr
37	Turn	High	11.50383, 77.40833	15 KM/Hr
38	Turn	High	11.50406, 77.40680	15 KM/Hr
39	Blind Spot	Blind Spot	11.50710, 77.40661	10 KM/Hr
40	Turn	Medium	11.50500, 77.39624	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
8	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
9	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
10	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium
11	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
12	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
13	hospital	Naarayani Hospital	11.4157401, 77.5773087	30 km/h	Medium
14	hospital	P K Hospital	11.4175923, 77.5640962	30 km/h	Medium
15	hospital	Dharun Hospital	11.4223638, 77.5597735	30 km/h	Medium
16	hospital	Dr. Ramaswamy Hospital	11.4249065, 77.5535658	30 km/h	Medium
17	hospital	Dakshin Trauma Centre	11.454593, 77.4461819	30 km/h	Medium
18	hospital	Dakshin Hospital	11.4546301, 77.445905	30 km/h	Medium
19	hospital	Sri Varahi Herbal Research Centre and Hospital	11.4539333, 77.4427438	30 km/h	Medium
20	clinic	Germen Homeopathy Clinic	11.4536305, 77.4394692	30 km/h	Medium
21	hospital	JP Hospital	11.454765, 77.438257	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	KKS Hospital	11.4551792, 77.4392927	30 km/h	Medium
24	hospital	Kumaran Hospital	11.4533971, 77.4349621	30 km/h	Medium
25	hospital	Siranjeeve Hospitals	11.453132, 77.4344382	30 km/h	Medium
26	clinic	Anandhan Clinic	11.4548037, 77.4349074	30 km/h	Medium
27	hospital	Ramachandran Hospital	11.452242, 77.434641	30 km/h	Medium
28	hospital	Dr. Rathinasamy Hospital	11.457562, 77.4327752	30 km/h	Medium
29	hospital	Gobi Neuro Mind Care	11.4548344, 77.434649	30 km/h	Medium
30	hospital	Hariharan Hospital	11.4523578, 77.4299287	30 km/h	Medium
31	hospital	Vasanth Hospital	11.4526864, 77.4292173	30 km/h	Medium
32	clinic	Jeeva Nursing Home	11.4533031, 77.4294145	30 km/h	Medium
33	hospital	Sathya Hospital	11.4561856, 77.4298694	30 km/h	Medium
34	hospital	S K Hospital	11.4540106, 77.430108	30 km/h	Medium
35	hospital	Sivakami Hospital	11.4531846, 77.4299288	30 km/h	Medium
36	clinic	Sri Sudha Physiotherapy Clinic	11.4559915, 77.4300217	30 km/h	Medium
37	hospital	Best Hospital	11.4543558, 77.4263617	30 km/h	Medium
38	clinic	Anand Nursing Home	11.4553169, 77.4296248	30 km/h	Medium
39	clinic	Kanmani Child Care Clinic	11.4545273, 77.428668	30 km/h	Medium
40	hospital	Rajasekaran Hospital	11.454463, 77.4262	30 km/h	Medium
41	hospital	Samuel Hospital	11.4542555, 77.4271194	30 km/h	Medium
42	hospital	Sivasakthi Hospital	11.4544711, 77.4297229	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
43	hospital	SS Children's Hospital	11.4555386, 77.4296357	30 km/h	Medium
44	clinic	Sri Dhanvantari Siddha Clinic	11.4561216, 77.4295797	30 km/h	Medium
45	hospital	Neha Hospital	11.4560138, 77.4270219	30 km/h	Medium
46	hospital	M.S Children's Hospital	11.4556012, 77.4301006	30 km/h	Medium
47	hospital	Gobi Diabetic Hospital	11.455394, 77.4301181	30 km/h	Medium
48	clinic	Dr. Ramesh Babu Clinic	11.4560871, 77.4296353	30 km/h	Medium
49	hospital	Sankareswari Eye Hospital	11.4572997, 77.428765	30 km/h	Medium
50	hospital	Arun Hospital	11.4548852, 77.4250314	30 km/h	Medium
53	hospital	Government Hospital	11.4565126, 77.4288768	30 km/h	Medium
54	hospital	Sri Chakra Hospital	11.4544464, 77.4228363	30 km/h	Medium
55	hospital	U M Hospital	11.4544151, 77.4240386	30 km/h	Medium
56	hospital	Dr. Gopal Hospital	11.455258, 77.422901	30 km/h	Medium
57	clinic	VG Ortho Centre	11.4550504, 77.4225102	30 km/h	Medium
58	clinic	Palaniappa Clinic	11.4555925, 77.4202828	30 km/h	Medium
59	hospital	Abi SK Hospital	11.457163, 77.412487	30 km/h	Medium
60	clinic	Nalam Vazh S.K Siddha and Varma Clinic	11.457576, 77.412164	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
7	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
23	school	Shree Vidyalaya Matric Higher Secondary School	11.4536114, 77.4383163	30 km/h	Medium
51	school	Diamond Jubilee Higher Secondary School	11.4529541, 77.4280306	30 km/h	Medium
52	school	CKK Palaniammal School	11.4569211, 77.4276828	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45509, 77.81383



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40046, 77.65371



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40035, 77.65383



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39829, 77.65233



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42079, 77.56127



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45790, 77.40551



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46345, 77.40571



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46390, 77.40581



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46411, 77.40610



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46407, 77.40677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46630, 77.40811



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47136, 77.41058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47169, 77.41050



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47207, 77.41009



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47242, 77.41006



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47535, 77.41069



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47796, 77.40971



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47858, 77.40888



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47910, 77.40859



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47932, 77.40859



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48060, 77.41086



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48081, 77.41092



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48178, 77.40992



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48370, 77.40914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48441, 77.40933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48861, 77.41164



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49474, 77.41231



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49523, 77.40994



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50383, 77.40833



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50406, 77.40680



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.50710, 77.40661



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50500, 77.39624

